Problem statement Send feedback

Your team has been working on an online e-commerce website at ABC Company. Suddenly, the website faces a large number of requests in a short period of time, exceeding the system capacity, which leads to Flooding. These flood attacks cause the system to receive too much traffic for the server to buffer, causing them to slow down and eventually stop. The application's target users are also experiencing errors about "too many connections" or something similar when visiting a website or using the app. You, along with the team, urgently need to solve this significant security issue. What should be the solution to this problem? Think about limiting the user requests and setting a limit. What will the advantages of such a system be?

Introduction & Advantages

eg 100 API Calls /min

- Resource Management
- against DoS (Denial of Service attacks)
- against malacious users etc
- Billing – how many users visiting.

Features

Functional :- → Response what if user exceeds API limit.

Non-functional :- Design Goals

Estimation :- No. of users

HLD :- Algo

Scaling

Problem statement

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What should be the Functional Requirement like limiting the user requests in a time window and Non-Functional requirements like availability or latency of the API Rate Limiter? You can also specify any extended requirement or feature that the system should meet. We now need to estimate the system capacity keeping in mind the performance constraints. You should also calculate the traffic estimations of the system, assuming that the average number of requests by users per second is 1000.